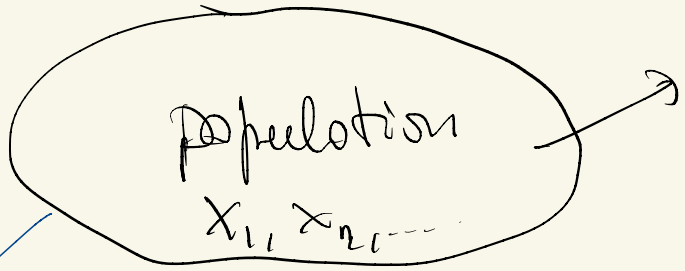
Eco 202 – Fall 2026

Class 14

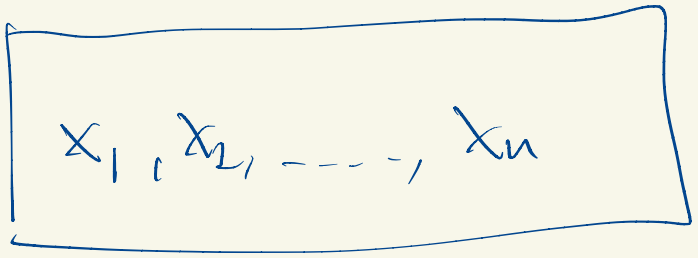
Review



parameter
(a non-random feature of the distribution of the data)

sample

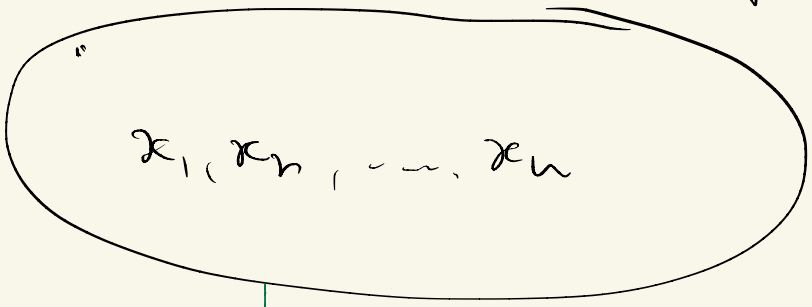
iid = independent and identically distributed



↳ $\mathbb{E}[x_i] = \mu$ for all i

$V[x_i] = \sigma^2$ for all i

we see a realization



← is your data!

↳ estimator: your "best" guess of what the parameter is

Eg. $\hat{\mu} = \frac{1}{n}\sum_{i=1}^{n} x_i$ → random

Estimators

- $\hat{\theta}$ is an unbiased estimator of a parameter θ if

$$\mathbb{E}[\hat{\theta}] = \theta$$

- $\mathbb{V}[\hat{\theta}]$ is the variance of the sampling distribution of $\hat{\theta}$

- $MSE[\hat{\theta}] = \mathbb{E}[(\hat{\theta} - \theta)^2]$

$$= \mathbb{V}[\hat{\theta}] + (\mathbb{E}[\hat{\theta}] - \theta)^2$$

Hypothesis testing (and next time confidence intervals)

- H_0 is the null hypothesis
- H_1 is the alternative hypothesis

} are mutually exclusive.

Example:

$$H_0: \mu = \mu_0 \quad \text{vs.} \quad H_1: \mu \neq \mu_0$$

Goal: to develop a rule, or a hypothesis testing procedure, to decide whether H_0 is compatible (or not) with my data.

($\mu > \mu_0$)

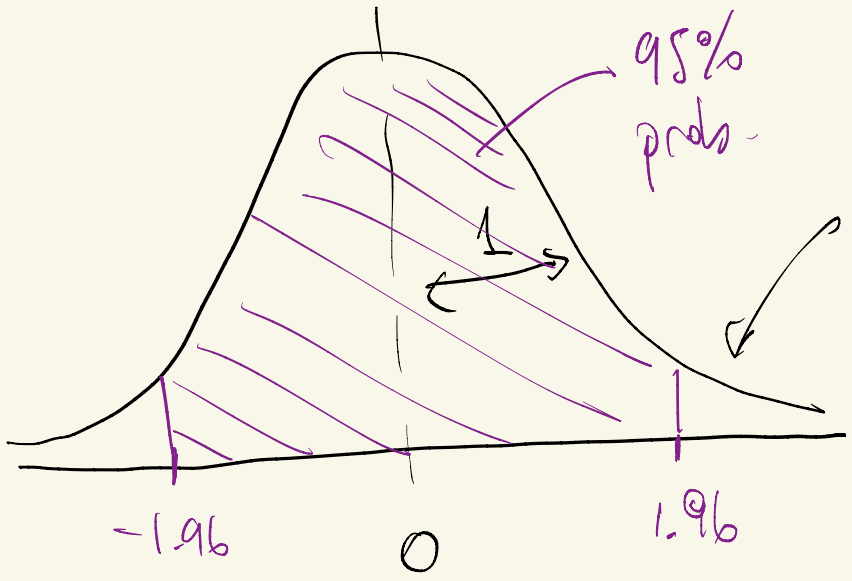
($\mu < \mu_0$)

- **Test statistic**: is a function of the data that we use to conduct a hypothesis test.

Example: $H_0: \mu = \mu_0$ vs. $H_1: \mu \neq \mu_0$

$$T = \frac{\bar{X}_n - \mu_0}{\sqrt{\mathbb{V}[\bar{X}_n]}} \longrightarrow \begin{cases} \mathbb{E}[T] = 0 \\ \mathbb{V}[T] = 1 \end{cases}$$

null distribution $\leftarrow$ { By the CLT, it is approximately standard Gaussian.



Standard Gaussian

$$T = \frac{\bar{X}_n - \mu_0}{\sqrt{\mathbb{V}[\bar{X}_n]}}$$

- **Rejection Rule**:

Reject H_0 if and only if $\underbrace{\left| \frac{\bar{X}_n - \mu_0}{\sqrt{\mathbb{V}[\bar{X}_n]}} \right|}_{T} > c$

c: critical value

Taxonomy of Errors

	H_0 true	H_0 not true
No reject H_0	✓	type II error lack of power
Reject H_0	type I error false discovery	✓

Rule: Reject H_0 iff $|T| > c$

T random!

- **Significance level**: maximum tolerance for type I error (false discovery)

$$\mathbb{P}[\text{type I error}]$$

$$= \mathbb{P}[\text{Rejecting } H_0 \mid H_0 \text{ is true}]$$

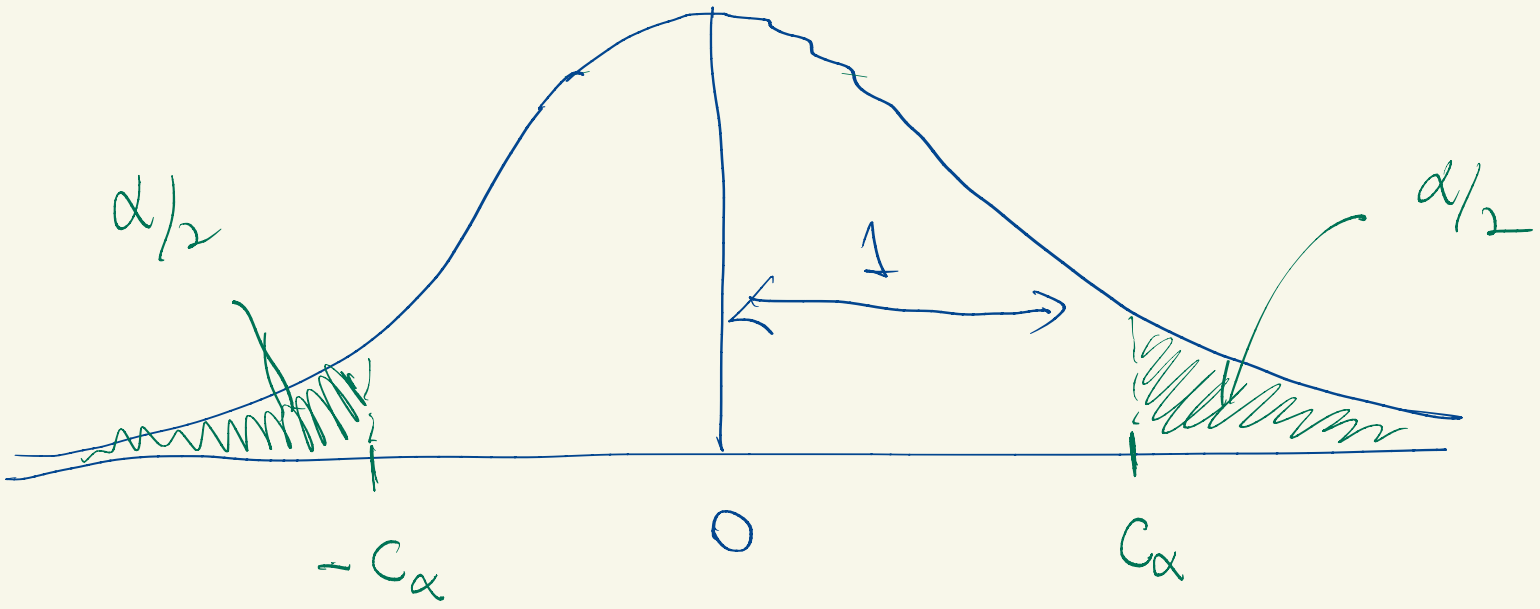
$$= \mathbb{P}[|T| > c \mid H_0 \text{ is true}]$$

$$= \mathbb{P}\left[\left|\frac{\bar{X}_n - \mu_0}{\sqrt{\mathbb{V}[\bar{X}_n]}}\right| > c \mid H_0 \text{ is true}\right] = \alpha$$

significance level!

$$\Rightarrow P\left[-c \leq \frac{\bar{X}_n - \mu_0}{\sqrt{V[\bar{X}_n]}} \leq c \,\middle|\, \text{Hoishua}\right] = \alpha$$

$$\frac{\bar{X}_n - \mu_0}{\sqrt{V[\bar{X}_n]}} \sim N(0,1)$$



$\alpha = 0.05 \Rightarrow C_\alpha \simeq 1.96$

$\alpha = 0.01 \Rightarrow C_\alpha \simeq 2.576$

Note:

Reject H_0 iff $\left| \frac{\bar{X}_n - \mu_0}{\sqrt{\mathbb{V}[\bar{X}_n]}} \right| > c_\alpha$

equivalent to

Reject H_0 iff $\bar{X}_n > \mu_0 + c_\alpha \sqrt{\mathbb{V}[\bar{X}_n]}$

or

$\bar{X}_n < \mu_0 - c_\alpha \sqrt{\mathbb{V}[\bar{X}_n]}$

$H_0 : \mu = \mu$ vs $H_1 : \mu > \mu_0$



$H_0 : \mu = \mu_0$ vs $H_1 : \mu < \mu_0$

